

THE WEIL PAIRING, ENDOMORPHISMS AND AUTOMORPHISMS

SANTIAGO ARANGO-PÍÑEROS

Three problems, each one filling a gap the notes leave open. Section 1 proves that the determinant of the mod N Galois representation is the cyclotomic character, which the notes only assert. Section 2 exhibits an endomorphism ring that is an order in a definite quaternion algebra, the third case of Theorem 4.7.8, on a single curve over $\mathbb{F}_9$. Section 3 computes the automorphism groups of order 12 and 24 in characteristic 3 and 2, the two cases of Theorem 4.7.11 that the notes do not prove, and they are exactly the two in which the group is not cyclic. Section 4 checks all three in Magma.

Reading. Sections 4.6 and 4.7 of the notes [AP26], in particular the definitions of a quaternion algebra $(a, b \mid \mathbb{Q})$, of its reduced trace and norm, and of an order (Definition 4.7.1 to Definition 4.7.6); and [Sil09, §§III.8–III.10], with [Sil09, Appendix A.1] for the automorphism groups in characteristic 2 and 3.

Standing notation. As in the notes, k is a field, $\bar{k}$ an algebraic closure and $k^{\text{sep}} \subseteq \bar{k}$ the separable closure. For $N \geq 2$ with $\text{char } k \nmid N$ we write

$$e_N: E[N](k^{\text{sep}}) \times E[N](k^{\text{sep}}) \rightarrow \mu_N(k^{\text{sep}})$$

for the Weil pairing, and

$$\bar{\rho}_{E,N}: \text{Gal}_k \rightarrow \text{Aut}(E[N](k^{\text{sep}})) \cong \text{GL}_2(\mathbb{Z}/N\mathbb{Z})$$

for the mod N Galois representation, the isomorphism depending on a choice of basis; $\rho_{E,\ell}$ is the ℓ -adic representation on $T_\ell(E)$. Also $[m]$ is multiplication by m , $\hat{\varphi}$ is the dual of an isogeny φ , and $F \in \text{End}(E)$ is the q -th power Frobenius endomorphism of a curve over $\mathbb{F}_q$. Beware that φ_ℓ already means the map induced by φ on $T_\ell(E)$, so the Frobenius *endomorphism* is F and never φ_q .

How to use this. Allow about 15 minutes for §1, 20 for §2 and 20 for §3. In Problem 1.2 the heart is part (3), and parts (4) and (5) are corollaries to finish at home. In Problem 2.3, parts (1)–(3) are the session and parts (4) and (5) the payoff; they can be started there and finished at home. In Problem 3.2, do characteristic 3 (parts (1)–(3)) in the session and characteristic 2 (parts (4) and (5)) at home: it is the same technique with more bookkeeping. Section 4 needs no time in the session; it is worked examples to read and re-run afterwards.

1. THE WEIL PAIRING AND THE CYCLOTOMIC CHARACTER

Fix $N \geq 2$ with $\text{char } k \nmid N$, so that $E[N](k^{\text{sep}}) \cong (\mathbb{Z}/N\mathbb{Z})^2$.

Fact 1.1 (Properties of the Weil pairing). By Proposition 4.6.3 of the notes, the Weil pairing e_N is

- (1) bilinear;
- (2) alternating: $e_N(T, T) = 1$, so $e_N(S, T) = e_N(T, S)^{-1}$;
- (3) non-degenerate: if $e_N(S, T) = 1$ for all $S \in E[N](k^{\text{sep}})$, then $T = O$;
- (4) Galois equivariant: $\sigma(e_N(S, T)) = e_N(\sigma S, \sigma T)$ for $\sigma \in \text{Gal}_k$;
- (5) compatible: $e_{NN'}(S, T) = e_N([N']S, T)$;
- (6) adjoint for isogenies: $e_N(\varphi S, T) = e_N(S, \hat{\varphi} T)$.

Problem 1.2 (The determinant is the cyclotomic character).

- (1) Let $\zeta \in \mu_N(k^{\text{sep}})$ be primitive. Show that for each $\sigma \in \text{Gal}_k$ there is a unique $\chi_N(\sigma) \in \mathbb{Z}/N\mathbb{Z}$ with $\sigma(\zeta) = \zeta^{\chi_N(\sigma)}$, that $\chi_N(\sigma)$ is a unit, and that the resulting map $\chi_N: \text{Gal}_k \rightarrow (\mathbb{Z}/N\mathbb{Z})^\times$ is a homomorphism independent of the choice of ζ . This is the **mod N cyclotomic character**.
- (2) Let S, T be a basis of $E[N](k^{\text{sep}})$ and put $\zeta := e_N(S, T)$. Show that ζ is a *primitive* N -th root of unity, using only parts (1)–(3) of Fact 1.1.
- (3) Write $\sigma S = aS + cT$ and $\sigma T = bS + dT$, so that $\bar{\rho}_{E,N}(\sigma) = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ in the basis S, T . Expand $e_N(\sigma S, \sigma T)$ and deduce

$$\det \bar{\rho}_{E,N} = \chi_N.$$

- (4) Take $k = \mathbb{Q}$. Show that $\det \bar{\rho}_{E,N}$ is surjective for *every* elliptic curve E over $\mathbb{Q}$, that $\mathbb{Q}(\zeta_N) \subseteq \mathbb{Q}(E[N])$, and that $\det \bar{\rho}_{E,N}(c) = -1$ for c a complex conjugation: the representation is **odd**.
- (5) Take $k = \mathbb{F}_q$ with q prime to N . Show that $\det \bar{\rho}_{E,N}(F) = q$, and check this against the identity $\det \varphi_\ell = \deg \varphi$ of **Proposition 4.6.7**.

Remark 1.3. Part (3) gives back **Corollary 4.6.5** of the notes: if $E[N](k^{\text{sep}}) \subseteq E(k)$, then $\bar{\rho}_{E,N}$ is trivial, so χ_N is trivial, so every $\sigma \in \text{Gal}_k$ fixes a primitive N -th root of unity, and $\mu_N(k^{\text{sep}}) \subseteq k$.

2. A QUATERNIONIC ENDOMORPHISM RING

Theorem 4.7.8 of the notes classifies $\text{End}(E)$ into three cases, $\mathbb{Z}$, an order in an imaginary quadratic field, or an order in a definite quaternion algebra over $\mathbb{Q}$, and gives examples of the first two only. Here is an example of the third, on the smallest curve that has one. Throughout this section

$$\bar{E}: y^2 = x^3 - x \quad \text{over } \mathbb{F}_3.$$

Fact 2.1 (Two results from the notes). Let E be an elliptic curve over $\mathbb{F}_q$ with F its q -power Frobenius and $a := q + 1 - \#E(\mathbb{F}_q)$.

- (1) F satisfies $F^2 - [a]F + [q] = 0$ in $\text{End}(E)$: by **Proposition 4.6.7** the characteristic polynomial of F_ℓ is $T^2 - aT + q$, and an endomorphism satisfies the characteristic polynomial of its action on $T_\ell(E)$.
- (2) If $\varphi: E_1 \rightarrow E_2$ is separable and $\psi: E_1 \rightarrow E_3$ is an isogeny with $\ker \varphi \subseteq \ker \psi$, then $\psi = \lambda \circ \varphi$ for a unique isogeny $\lambda: E_2 \rightarrow E_3$ (**Proposition 4.3.9**).

Fact 2.2 (Two facts about $(-1, -3 \mid \mathbb{Q})$). Let $D := (-1, -3 \mid \mathbb{Q})$, the quaternion algebra with $i^2 = -1$, $j^2 = -3$ and $ij = -ji$ (**Definition 4.7.1**).

- (1) $D \otimes_{\mathbb{Q}} \mathbb{Q}_p$ is split for every prime $p \neq 3$, and $D \otimes_{\mathbb{Q}} \mathbb{Q}_3$ is not; so D is a division algebra, and its reduced discriminant, the product of the primes at which it is not split, is 3 [Voi21, Exercise 11.11 and §14.1].
- (2) An order $\mathcal{O} \subseteq D$ is maximal if and only if $\text{discrd}(\mathcal{O}) = 3$, where $\text{discrd}(\mathcal{O})^2 = |\det(\text{trd}(\alpha_k \alpha_l))_{k,l}|$ for a $\mathbb{Z}$ -basis $\alpha_1, \dots, \alpha_4$ of $\mathcal{O}$; and if $\mathcal{O} \subseteq \mathcal{O}'$, then $\text{discrd}(\mathcal{O}) = [\mathcal{O}' : \mathcal{O}] \text{discrd}(\mathcal{O}')$ [Voi21, Theorem 15.5.5]; see the end of **Section 4.7.1** of the notes.

Problem 2.3 ($\text{End}(\bar{E}_{\mathbb{F}_3})$ is a maximal order in a quaternion algebra).

- (1) Check that $\bar{E}$ is an elliptic curve over $\mathbb{F}_3$, list $\bar{E}(\mathbb{F}_3)$, and deduce that $a = 0$, that $\bar{E}$ is supersingular, and that

$$F^2 = [-3].$$

Record also that every point of $\bar{E}[2](\bar{\mathbb{F}}_3)$ is defined over $\mathbb{F}_3$: this is what makes part (5) work.

- (2) Show that $x^2 + 1$ is irreducible over $\mathbb{F}_3$, so that $\mathbb{F}_9 = \mathbb{F}_3(i)$ with $i^2 = -1$. Show that

$$[i](x, y) := (-x, iy)$$

defines an endomorphism of $\bar{E}_{\mathbb{F}_9}$, and that $[i]^2 = [-1]$. Why is a morphism $\bar{E} \rightarrow \bar{E}$ fixing ∞ automatically a group homomorphism?

- (3) Show that $F[i] = -[i]F$. Conclude that $\text{End}(\bar{E}_{\mathbb{F}_9})$ is not commutative, so that $\bar{E}_{\mathbb{F}_9}$ falls into the third case of the classification.

- (4) Deduce that

$$\text{End}(\bar{E}_{\mathbb{F}_9}) \otimes_{\mathbb{Z}} \mathbb{Q} \cong D = (-1, -3 \mid \mathbb{Q}).$$

Write down the reduced norm of D and check that D is definite ([Proposition 4.7.5](#)); how does the reduced norm compare with the degree of an endomorphism?

- (5) Show that $\frac{1+F}{2}$ and $\frac{i+iF}{2}$ lie in $\text{End}(\bar{E}_{\mathbb{F}_9})$. *Hint: use (2) of Fact 2.1 with $\varphi = [2]$.* Then show that

$$\mathcal{O} := \mathbb{Z} + \mathbb{Z}i + \mathbb{Z} \frac{1+F}{2} + \mathbb{Z} \frac{i+iF}{2}$$

is an order of D ([Definition 4.7.6](#)), compute $\text{discrd}(\mathcal{O})$, and conclude with Fact 2.2 that $\text{End}(\bar{E}_{\mathbb{F}_9}) = \mathcal{O}$ is a maximal order.

3. AUTOMORPHISMS IN CHARACTERISTIC 2 AND 3

Fact 3.1 (The automorphism group, from the notes). Let E be an elliptic curve over k . By [Theorem 4.7.11](#), $\text{Aut}_{\bar{k}}(E)$ is finite, of order

$$\# \text{Aut}_{\bar{k}}(E) = \begin{cases} 2 & \text{if } j(E) \neq 0, 1728, \\ 4 & \text{if } j(E) = 1728 \text{ and } \text{char } k \neq 2, 3, \\ 6 & \text{if } j(E) = 0 \text{ and } \text{char } k \neq 2, 3, \\ 12 & \text{if } j(E) = 0 = 1728 \text{ and } \text{char } k = 3, \\ 24 & \text{if } j(E) = 0 = 1728 \text{ and } \text{char } k = 2. \end{cases}$$

If $\text{char } k \neq 2, 3$, then $\text{Aut}_{k^{\text{sep}}}(E) \cong \mu_n(k^{\text{sep}})$ with $n = \# \text{Aut}_{\bar{k}}(E)$, via $[\zeta](x, y) := (\zeta^2 x, \zeta^3 y)$. The notes prove this for $\text{char } k \neq 2, 3$ only.

Recall the general change of variables preserving a Weierstrass equation ([Theorem 4.1.5](#)):

$$(3.1) \quad x = u^2 X + r, \quad y = u^3 Y + u^2 s X + t, \quad u \in \bar{k}^\times, \quad r, s, t \in \bar{k},$$

under which the coefficients transform by

$$(3.2) \quad \begin{aligned} ua'_1 &= a_1 + 2s, \\ u^2 a'_2 &= a_2 - sa_1 + 3r - s^2, \\ u^3 a'_3 &= a_3 + ra_1 + 2t, \\ u^4 a'_4 &= a_4 - sa_3 + 2ra_2 - (t + rs)a_1 + 3r^2 - 2st, \\ u^6 a'_6 &= a_6 + ra_4 + r^2 a_2 + r^3 - ta_3 - t^2 - rta_1. \end{aligned}$$

The whole point of this section is visible in (3.2): the parameters s and r are killed by the terms $2s$ and $3r$, and those terms vanish in characteristic 2 and 3 respectively.

Problem 3.2 (The two non-cyclic cases).

- (1) Let $\text{char } k = 3$. Show that every short Weierstrass equation $y^2 = x^3 + Ax + B$ has $j = 0$, and explain why a curve with $j \neq 0$ in characteristic 3 needs an a_2x^2 term. So the case “ $j = 0 = 1728$, $\text{char } k = 3$ ” of Fact 3.1 is the short-Weierstrass, supersingular one.
- (2) Let $\text{char } k = 3$ and $E: y^2 = x^3 - x$. Using (3.2), show first that $s = t = 0$, and then that (3.1) is an automorphism of E if and only if

$$u^4 = 1 \quad \text{and} \quad r^3 - r = 0,$$

i.e. $u \in \mu_4 \subseteq \mathbb{F}_9^\times$ and $r \in \mathbb{F}_3$. Conclude $\#\text{Aut}_{\mathbb{F}_3}(E) = 12$, and that all 12 are already defined over $\mathbb{F}_9$. Where did r become free, and what would have forced $r = 0$ if $\text{char } k \neq 3$?

- (3) Write $\varphi_{u,r}(x, y) := (u^2x + r, u^3y)$. Compute $\varphi_{u',r'} \circ \varphi_{u,r}$ and deduce

$$\text{Aut}_{\mathbb{F}_3}(E) \cong \mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z},$$

with $\mathbb{Z}/4\mathbb{Z}$ acting through its quotient $\{\pm 1\}$. Show this group has a *unique* element of order 2, and deduce that it is neither cyclic, nor dihedral, nor A_4 : it is the dicyclic group of order 12. Which automorphism is the unique involution?

- (4) Let $\text{char } k = 2$ and $E: y^2 + y = x^3$. Using (3.2), show that $a'_1 = a_1$ automatically, that $a'_2 = 0$ forces $r = s^2$, and then that (3.1) is an automorphism if and only if

$$u^3 = 1, \quad s^4 = s, \quad t^2 + t = s^6.$$

Deduce that $s \in \mathbb{F}_4$, count the solutions, and conclude $\#\text{Aut}_{\mathbb{F}_2}(E) = 24$, all defined over $\mathbb{F}_4$.

- (5) Write $\psi_{s,t}(x, y) := (x + s^2, y + sx + t)$ for the automorphisms with $u = 1$. Show that

$$\psi_{s',t'} \circ \psi_{s,t} = \psi_{s+s', t+t'+s's^2}, \quad \psi_{s,t}^2 = \psi_{0,s^3},$$

and deduce that these 8 automorphisms form a copy of the quaternion group Q_8 . Conclude that

$$\text{Aut}_{\mathbb{F}_2}(E) \cong Q_8 \rtimes \mathbb{Z}/3\mathbb{Z} \cong \text{SL}_2(\mathbb{F}_3).$$

Remark 3.3 (Both groups are unit groups). Automorphisms are the units of the endomorphism ring, $\text{Aut}_{\bar{k}}(E) = \text{End}_{\bar{k}}(E)^\times$. In characteristic 2 and 3 the curve with $j = 0$ is supersingular, so by Remark 4.7.12 of the notes $\text{End}_{\bar{k}}(E)$ is a maximal order in the quaternion algebra over $\mathbb{Q}$ that is not split exactly at p and ∞ . For $p = 3$ that order is the one computed in Problem 2.3(5), and its unit group is the dicyclic group of order 12 of part (3). For $p = 2$ it is the Hurwitz order $\mathbb{Z} + \mathbb{Z}i + \mathbb{Z}j + \mathbb{Z}\frac{1+i+j+ij}{2}$ in $(-1, -1 \mid \mathbb{Q})$ [Voi21, §11.1], whose unit group is the binary tetrahedral group of order 24 of part (5). This is the content of the sentence in the notes after Theorem 4.7.11: when $\text{End}_{\bar{k}}(E)$ is commutative it is $\mathbb{Z}$ or an imaginary quadratic order, whose unit groups are the cyclic groups μ_2, μ_4, μ_6 , giving the first three lines of Fact 3.1, while a non-commutative $\text{End}_{\bar{k}}(E)$ is a quaternion order, and its unit group need not be, and here is not, cyclic.

4. MAGMA

Worked examples to read and re-run. Each one checks a problem above; the second and third check them essentially completely.

Example 4.1 (The cyclotomic character, numerically). Problem 1.2(5) over a finite field is something Magma can confirm outright: build a basis of $E[3]$ over the field where it becomes rational, apply the q -power Frobenius coordinatewise, and compare $e_3(FS, FT)$ with $e_3(S, T)^q$.

Choosing the example is itself instructive. We want $q \not\equiv 1 \pmod{3}$, so that χ_3 is *not* trivial and the check has content; by Remark 1.3 that already forces $E[3] \not\subseteq E(\mathbb{F}_q)$. Take $q = 5$ and E supersingular, so $a = 0$ and $E(\mathbb{F}_{25}) \cong (\mathbb{Z}/6\mathbb{Z})^2$ contains all of $E[3]$.

```

q := 5; N := 3;
E := EllipticCurve([GF(q) | 0, 1]); // y^2 = x^3 + 1
#E, IsSupersingular(E); // 6 true (so a = 0)
Ek := ChangeRing(E, GF(q^2));
#Ek, AbelianGroup(Ek); // 36, Z/6 + Z/6
T3 := [P : P in Points(Ek) | N*P eq Ek!0];
#T3; // 9
S := [P : P in T3 | P ne Ek!0][1];
T := [P : P in T3 | P ne Ek!0 and P ne S and P ne -S][1];
z := WeilPairing(S, T, N); Order(z); // 3: a primitive cube root
frob := func<P | P eq Ek!0 select Ek!0 else Ek![P[1]^q, P[2]^q]>;
WeilPairing(frob(S), frob(T), N) eq z^q; // true

```

The pairing $e_3(S, T)$ has order exactly 3, as Problem 1.2(2) predicts, and the last line is exactly $\det \bar{\rho}_{E,3}(F) = q$. Writing out the matrix of F in the basis S, T makes the determinant visible:

```

[ <a, b> : a, b in [0..2] | a*S + b*T eq frob(S) ]; // [ <1, 0> ]
[ <a, b> : a, b in [0..2] | a*S + b*T eq frob(T) ]; // [ <0, 2> ]
GF(3)!q; // 2

```

Here $FS = S$ and $FT = [2]T$, so the matrix is $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$, with determinant $2 = q \bmod 3$; another basis gives another matrix, never another determinant.

Example 4.2 (The endomorphism ring). All of Problem 2.3 can be checked in Magma. The endomorphisms $[i]$ and F are small enough to write as maps on points and verify pointwise, and the quaternion machinery then confirms that the order is maximal.

```

E3 := EllipticCurve([GF(3) | -1, 0]); // prints as y^2 = x^3 + 2*x
Points(E3);
// {0 (0 : 1 : 0), (0 : 0 : 1), (1 : 0 : 1), (2 : 0 : 1) 0}
#E3, 3 + 1 - #E3; // 4 0 (so a = 0)
IsSupersingular(E3), jInvariant(E3); // true 0

```

Warning 4.3. $\text{GF}(9)$ does *not* come with i . Magma's default generator of $\mathbb{F}_9$ is a root of the Conway polynomial $x^2 + 2x + 2$, so writing $\text{F9}<i> := \text{GF}(9)$ gives an i with $i^2 = i + 1$, not $i^2 = -1$, and every check below would silently fail. Construct the square root of -1 explicitly.

```

F9 := GF(9); i := Sqrt(F9!(-1));
i^2, MinimalPolynomial(i); // 2 (x^2 + 1), and 2 = -1
E9 := ChangeRing(E3, F9); #E9; // 16: the eigenvalues of F are +-sqrt(-3)
imap := func<P | P eq E9!0 select E9!0 else E9![-P[1], i*P[2]]>;
f3 := func<P | P eq E9!0 select E9!0 else E9![P[1]^3, P[2]^3]>;
forall{P : P in Points(E9) | imap(P) in Points(E9)}; // [i] is well defined
forall{<P,Q> : P, Q in Points(E9) | imap(P+Q) eq imap(P)+imap(Q)}; // additive
forall{P : P in Points(E9) | imap(imap(P)) eq -P}; // [i]^2 = [-1]
forall{P : P in Points(E9) | f3(f3(P)) eq -3*P}; // F^2 = [-3]
forall{P : P in Points(E9) | f3(imap(P)) eq -imap(f3(P))}; // F[i] = -[i]F
// all true

```

Now the algebra and the order. Magma's `QuaternionOrder` takes a $\mathbb{Z}$ -basis, so the four elements of Problem 2.3(5) can be handed to it directly; `Discriminant` of an order is its reduced discriminant.

```

A<ii,jj> := QuaternionAlgebra<Rationals() | -1, -3>;
RamifiedPrimes(A), Discriminant(A);           // [ 3 ] 3
O := QuaternionOrder([A!1, ii, (1+jj)/2, (ii+ii*jj)/2]);
Discriminant(O);                             // 3
IsMaximal(O);                                // true

```

The reduced discriminant is 3, the product of the primes at which D is not split, so the order is maximal and the inclusion of Problem 2.3(5) is an equality.

Example 4.4 (The two automorphism groups). Problem 3.2 too can be settled in Magma, including the group identifications, which are the fiddly part by hand. `AutomorphismGroup` returns a polycyclic group, and `IdentifyGroup` gives its label in the small-groups database.

```

E9 := ChangeRing(EllipticCurve([GF(3) | -1, 0]), GF(9));
G3 := AutomorphismGroup(E9); #G3;           // 12
E4 := ChangeRing(EllipticCurve([GF(2) | 0,0,1,0,0]), GF(4)); // y^2 + y = x^3
G2 := AutomorphismGroup(E4); #G2;           // 24
IdentifyGroup(G3), GroupName(G3);           // <12, 1> C3:C4
IdentifyGroup(G2), GroupName(G2);           // <24, 3> SL(2,3)
IsAbelian(G3), IsCyclic(G3);               // false false
{* Order(g) : g in G3 *};                  // {* 1, 2, 3^~2, 4^~6, 6^~2 *}
{* Order(g) : g in G2 *};                  // {* 1, 2, 3^~8, 4^~6, 6^~8 *}

```

Label $\langle 12, 1 \rangle$ is $\mathbb{Z}/3\mathbb{Z} \rtimes \mathbb{Z}/4\mathbb{Z}$ and $\langle 24, 3 \rangle$ is $\mathrm{SL}_2(\mathbb{F}_3)$, as computed by hand in (3) and (5). The multisets of element orders confirm the structural point of both parts: in each group there is exactly *one* element of order 2, which is what ruled out the dihedral group and A_4 in (3), and what pinned down Q_8 in (5).

The same two groups appear as unit groups of the maximal quaternion orders, which is Remark 3.3.

```

for d in [-1, -3] do
  Ad := QuaternionAlgebra<Rationals() | -1, d>;
  Od := MaximalOrder(Ad);
  print RamifiedPrimes(Ad), #UnitGroup(Od);
end for;
// [ 2 ] 24      <- Hurwitz order, characteristic 2
// [ 3 ] 12      <- the order of Problem 2.3, characteristic 3

```

So the 24 and the 12 of Fact 3.1 are the unit counts of the two maximal quaternion orders, computed without reference to any elliptic curve.

Looking ahead. The supersingular curves of section 2 are the ones whose Frobenius has $a = 0$; Hasse's bound and the structure of End over $\mathbb{F}_q$ are Chapter 5. In characteristic 0 the third case of the classification never occurs, and $\mathrm{End}_{\bar{k}}(E) \neq \mathbb{Z}$ becomes complex multiplication: Chapter 11. The oddness of $\rho_{E,\ell}$ from Problem 1.2(4) is the first hint of modularity.

REFERENCES

- [AP26] Santiago Arango-Piñeros, *The arithmetic geometry of elliptic curves*, 2026, <https://sarangop1728.github.io/teaching/714-fall-26/notes/>. Lecture notes for Math 714, UMass Amherst, Fall 2026. $\uparrow$
- [Sil09] Joseph H. Silverman, *The arithmetic of elliptic curves*, Second, Graduate Texts in Mathematics, vol. 106, Springer, Dordrecht, 2009. MR2514094 $\uparrow$

[Voi21] John Voight, *Quaternion algebras*, Graduate Texts in Mathematics, vol. 288, Springer, Cham, 2021.
MR4279905 ↑